School of Computer Science and Artificial Intelligence

Lab Assignment - 22.3

Course Name : AI Assistant Coding
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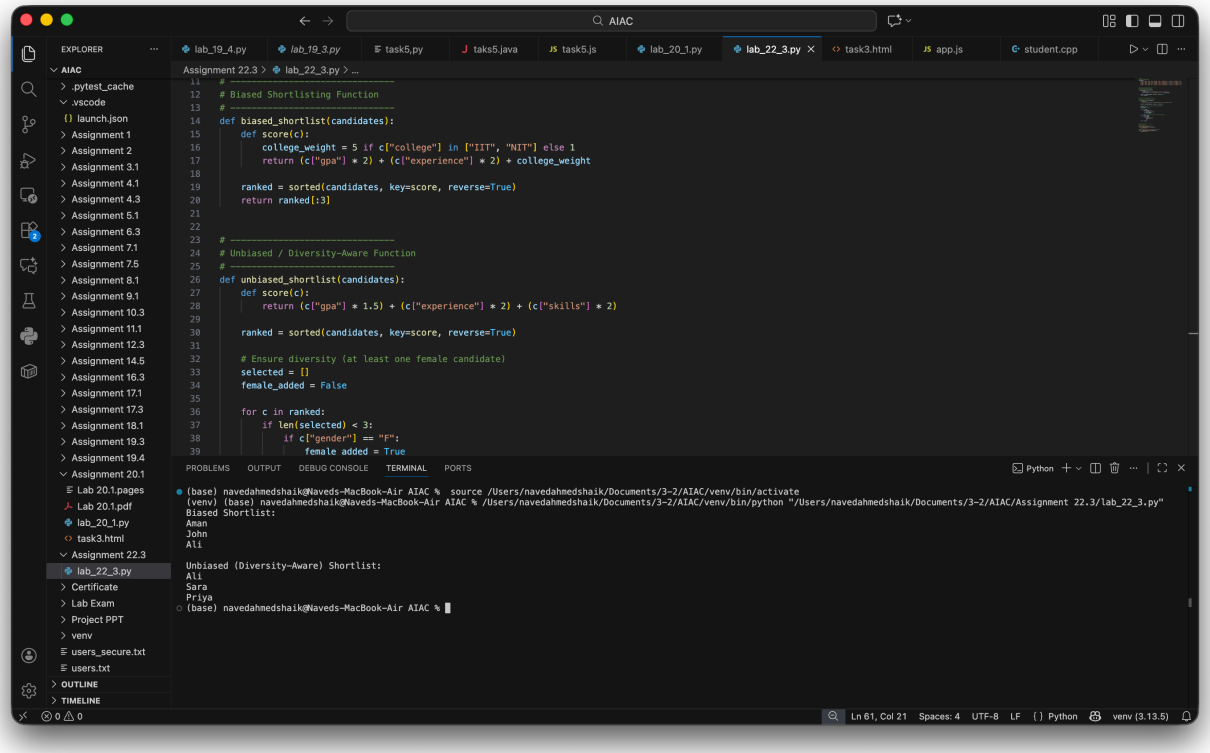
Batch No : 47-B

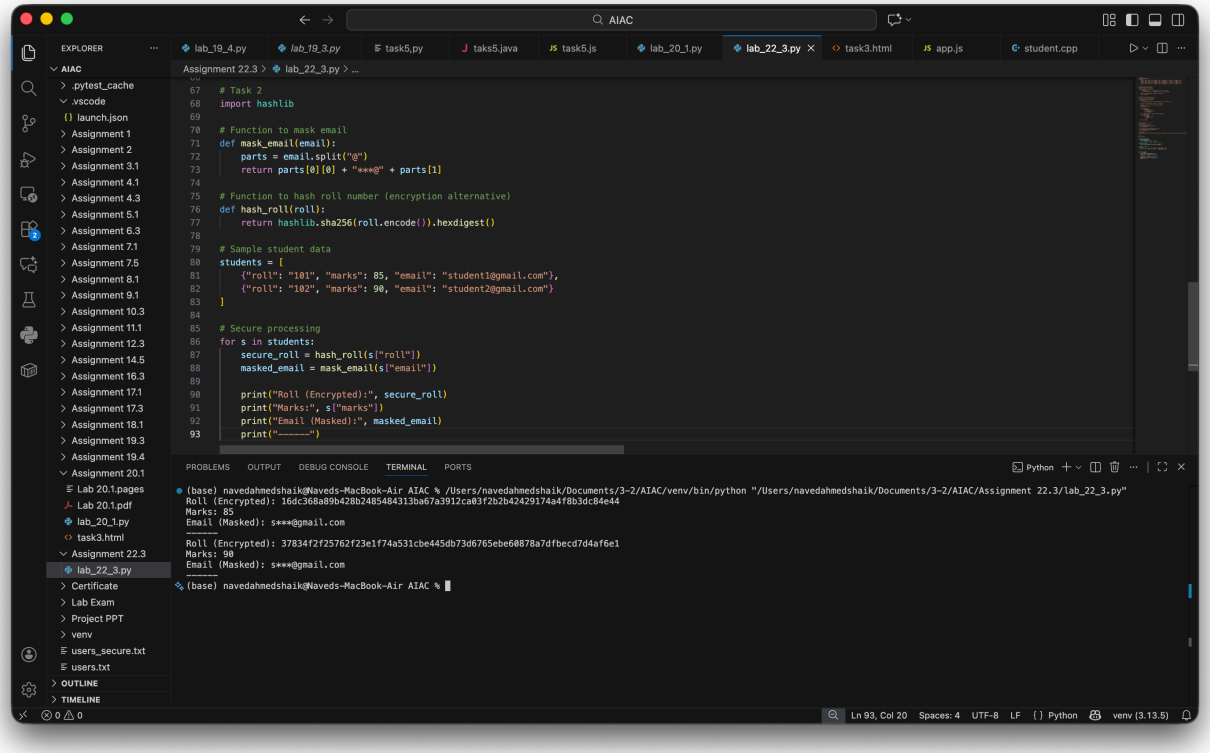
Task 1 – Bias Detection in AI Suggestions (Alternative Example)
Example Scenario:
Ask AI to generate a resume shortlisting algorithm multiple times.
• Change prompt wording (e.g., "select best candidate" vs "ensure diversity").
• Compare outputs and logic.
Discussion Points:
• Does the AI favor certain qualifications, backgrounds, or experiences?
• How could biased shortlisting algorithms affect equal employment opportunities?



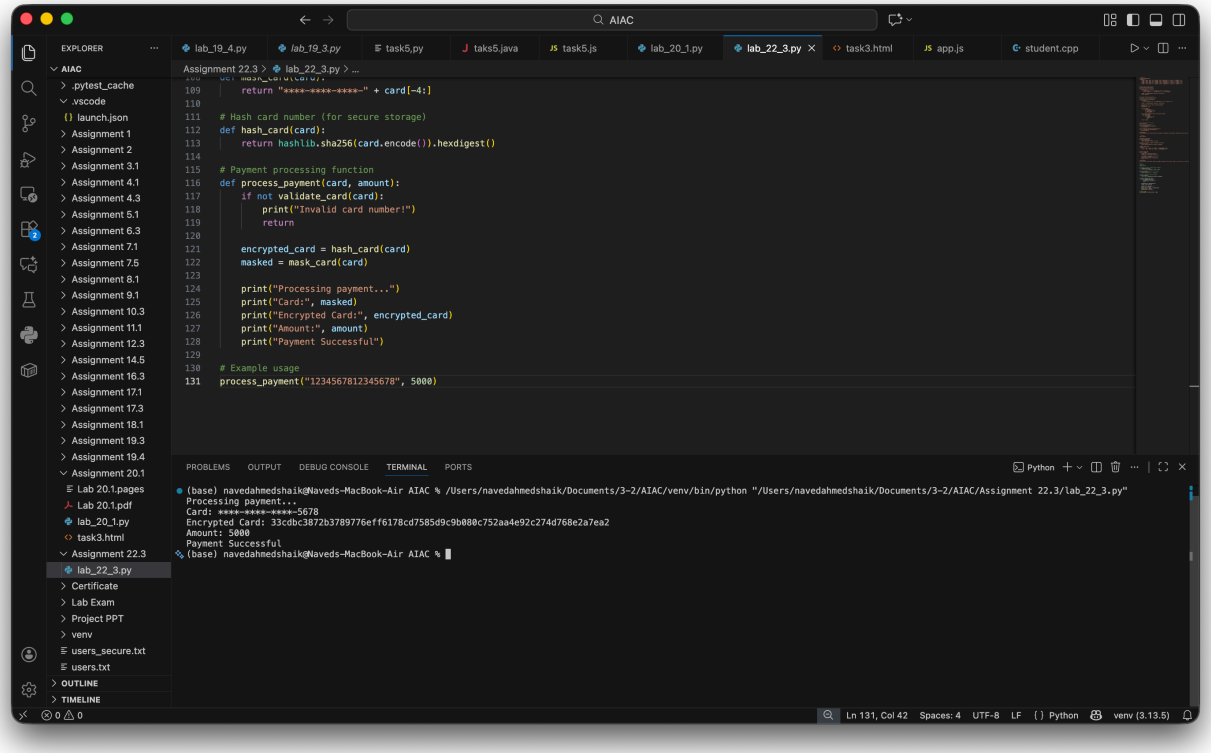
Explanation:
The biased_shortlist function gives extra weight to candidates from prestigious colleges, which can lead to overlooking talented candidates from less-known institutions. The unbiased_shortlist function focuses on a more holistic evaluation of candidates and ensures diversity by including at least one female candidate, thus promoting fairness and inclusivity in the selection process.

Task 2 – Privacy and Sensitive Data Handling (Alternative Example)
Example Scenario:
Ask AI to write code for processing student academic records (roll number, marks, email).
Discussion Points:
• How should student identity data be masked or encrypted?
• What privacy laws (e.g., FERPA, GDPR-like principles) are relevant?
• What ethical responsibility does the programmer have when using AI-generated code?

Explanation:
The original script likely stored roll numbers and emails in plain text, which is a security risk. In the improved version, roll numbers are hashed using SHA-256, making them unreadable and secure. Emails are masked to protect student privacy while still allowing for identification. This approach enhances data security and privacy, reducing the risk of sensitive information being exposed.

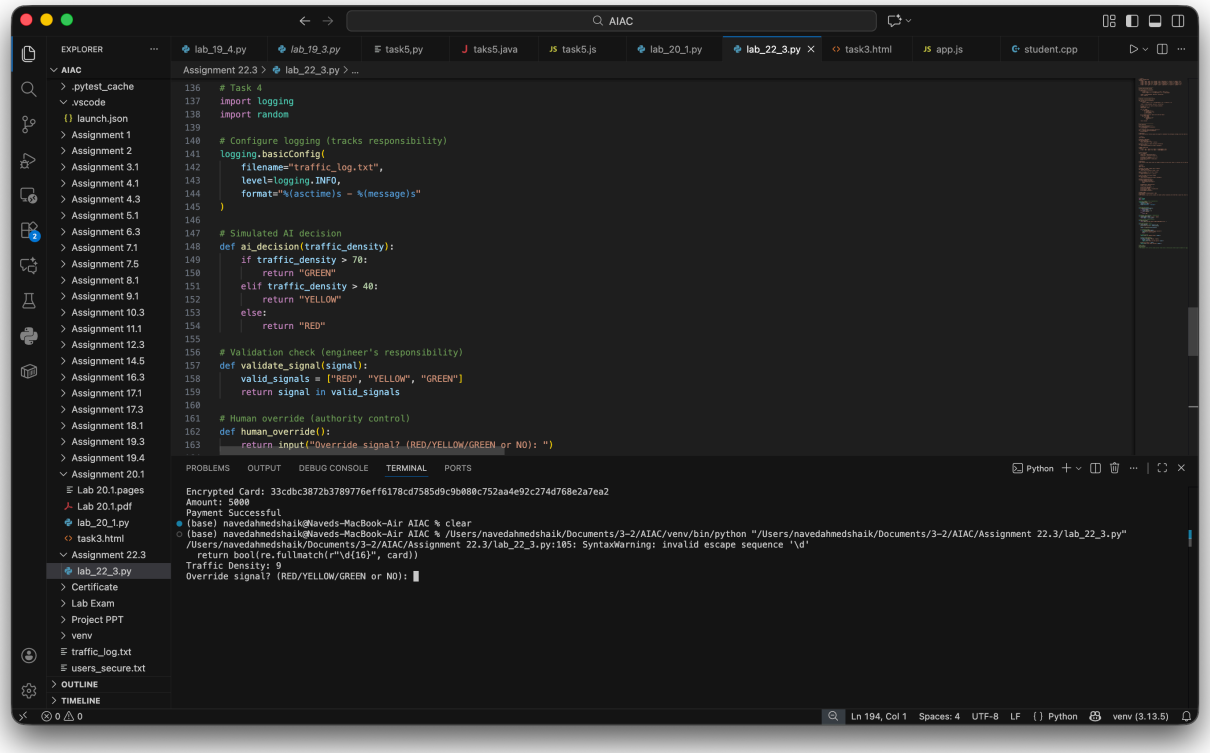


Task 3 – AI-Generated Security Risks (Alternative Example)
Example Scenario:
Use AI to generate a payment processing script for an online shopping system.
Student Tasks:
- **Identify risks such as:**
  - **Plain-text card data storage**
  - **Missing input validation**
- **Propose ethical coding standards to prevent financial fraud and misuse.**



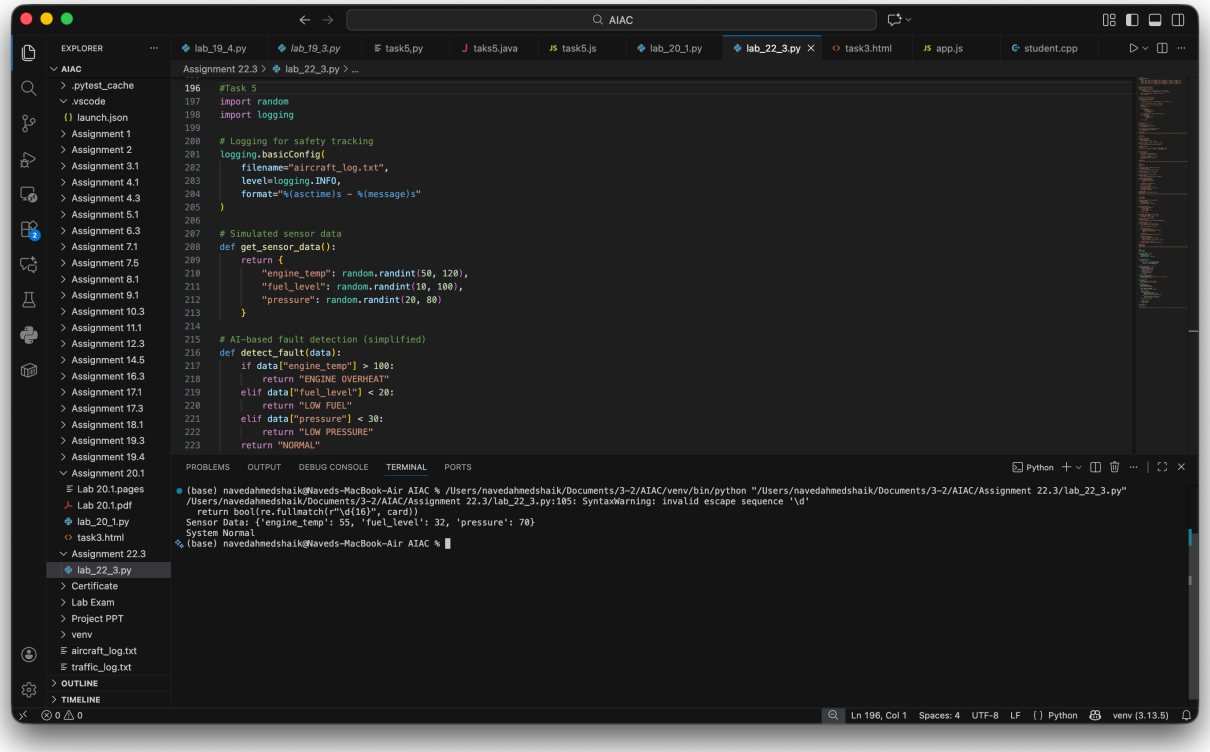
Explanation:
The original script may have accepted card numbers without validation and stored them in plain text, which is a major security vulnerability. The improved version includes validation to ensure the card number is in the correct format, masks the card number for display to protect user privacy, and hashes the card number for secure storage. This approach significantly enhances the security of sensitive payment information, reducing the risk of data breaches and unauthorized access.

Task 4 – Accountability in Human–AI Collaboration (Alternative Example)
Scenario:
AI generates code for an autonomous traffic signal control system.
Discussion Questions:
• If a software error causes accidents, who is accountable?
• Should responsibility lie with:
o AI tool developers
o Software engineers
o Government authorities deploying the system?
Students submit a short reflection on accountability.



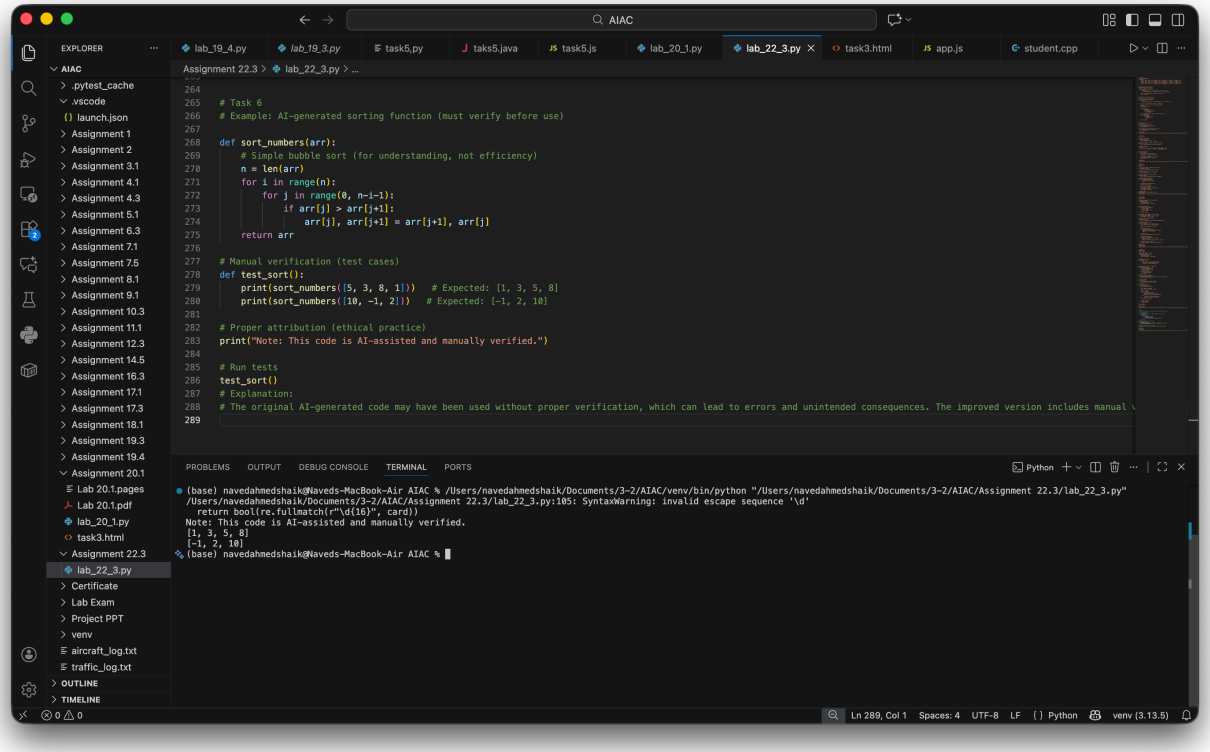
Explanation:
The original traffic control system may have relied solely on AI decisions without proper validation or logging, which can lead to unsafe traffic signals and lack of accountability. The improved version includes validation to ensure the AI's output is valid, logging to track decisions and overrides for accountability, and a human override option to allow for authority control in case of AI errors. This approach enhances the safety and reliability of the traffic control system while ensuring that responsibility is clearly defined and traceable.

Task 5 – Limitations of AI in Critical Systems (Alternative Example)
Example Scenario:
Ask AI to generate code for an aircraft maintenance fault detection system.
Discussion Points:
- **Can AI-generated code be trusted without expert validation?**
- **What human oversight is essential in aviation software?**
- **How should safety-critical code be reviewed and certified?**



Explanation:
The original aircraft system may have relied solely on AI for fault detection without human validation or fail-safe mechanisms, which can lead to catastrophic failures if the AI makes an error. The improved version includes human validation to confirm AI-detected faults, logging for accountability and tracking, and a fail-safe mechanism to ensure safety in case of AI errors. This approach enhances the reliability and safety of the aircraft system while ensuring that responsibility is clearly defined and that there are safeguards in place to prevent accidents.

Task 6 – Ethical Use of AI-Generated Code (Alternative Example)
Example Scenario:
Students analyze cases where AI-generated code is used in academic assignments or startups.
Discussion Points:
• Risks of submitting AI code without understanding it
• Copyright and license violations
Best practices:
o Proper citation
o Manual verification
o Ethical reuse of AI outputs



Explanation:
The original AI-generated code may have been used without proper verification, which can lead to errors and unintended consequences. The improved version includes manual verification through test cases to ensure the correctness of the sorting function. Additionally, it includes a note about the AI-assisted nature of the code, promoting transparency and ethical practice in software development. This approach ensures that the code is reliable and that the contributions of AI are acknowledged while maintaining human oversight for quality assurance.